# Cortical Decoders Trained on Other Individuals Age Slower Than Your Own

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## Abstract

A decoder fit to a singular animal's cortex drifts, losing accuracy over days. Usually the remedy is recalibrating on fresh data from the same subject. We report a different kind of robustness: a decoder built from one other individual, or pooled from a set of other individuals, ages more slowly than the subject's own. On the open BraiDyn-BC cohort (25 mice, 44 Allen-atlas cortical parcels, a $\sim$15-day operant protocol) we decode four task events and measure, per mouse, the AUC lost over two weeks by its own early-week decoder against one pooled over the other 24 mice and one built from a single other mouse. The personal decoder ages more by $+0.017$ AUC ($p = 5 \times 10^{-5}$, 73 of 100 mouse–event pairs). This also is not a data-volume effect: at a matched training-event count a decoder from a single other mouse still ages less. Hence we conclude this is rather an effect of not overfitting an individual's drift-prone idiosyncrasies. The effect holds past this cohort and recording modality. On an independent two-photon dataset from a different laboratory – Allen Visual Behavior, 23 mice, single-plane VISp populations decoding a stimulus change – the same session-held-out protocol gives $+0.023$ AUC ($p = 0.003$, 74% of mice), and the count-matched control reproduces there as well (self $-$ one other mouse $= +0.011$ AUC). The early-week accuracy cost of decoding across individuals is repaid as late-week stability, which suggests population priors as an alternative to per-subject recalibration. Code and a streamed-feature pipeline are released.

## 1 Introduction

Representational drift, the reorganization of an animal's neural code over days with fixed behavior, is now documented across sensory, motor, and association cortex and in the hippocampus [18, 3, 20]. This has a stark impact on neural decoding: a decoder trained on an animal today degrades on that same animal weeks later. The dominant response is recalibration, periodically refitting the decoder on fresh within-subject data [2, 5], which requires ongoing labelled data from the individual and treats each subject in isolation.

The code's cross-individual structure offers an alternate route to temporal robustness. Cortex-wide behavioral representations are considerably shared among individuals [11, 7, 8], which already allows models to decode behavior from unseen individuals [21]. The question we pose is temporal: does a decoder built from a population of other individuals resist drift better than a decoder built from the subject itself? This prediction, to our knowledge, has not been examined.

We test it on BraiDyn-BC [6], a widefield calcium-imaging resource whose $\sim$15-day operant protocol and shared 44-region Allen parcellation make within-subject and cross-subject decoders directly comparable across time, and replicate it on the Allen Visual Behavior two-photon cohort [1]. Our contributions:

- **Pooling resists drift.** Across a two-week period the population-pooled decoder loses less accuracy than the individual's own, with a paired asymmetry of $+0.017$ AUC ($p = 5 \times 10^{-5}$, 73 of 100 pairs; Section 4.1). The effect varies among the four events, but those are chained within a trial, so we treat the pooled estimate as the result and leave the per-event ranking uninterpreted (Section 4.2).

- **It is not a data-volume artifact.** At a matched training-event count a decoder trained on one other mouse ages less than one trained on the subject's own, by $+0.010$ AUC on average (Section 4.4).

- **It replicates on an independent dataset.** On the Allen Visual Behavior two-photon cohort the same protocol gives $+0.023$ AUC ($p = 0.003$, 74% of 23 mice), with the count-matched ordering also reproduced (Section 4.6).

- **Session-level validation is load-bearing.** A trial-level split inflates the personal decoder but not the pooled one, one-sidedly inflating the asymmetry; we recommend session-held-out validation as a default (Section 4.5).

## 2 Related work

Drift is characterized within individual subjects [18, 3, 20, 16], and most approaches to mitigating it also operate within the subject: manifold stabilization [2], self-calibrating recalibration [5, 15], and alignment to the subject's own reference day [4, 14]. Each uses later data from the same subject. Cross-subject structure is separately known to support readout [9, 19, 21] and to reduce overfitting when pooled [13, 12], and latent dynamics are preserved across animals to the point that a decoder transfers between them [17]. That conservation is why pooling is possible here; it does not imply pooling will *resist drift*, as a code can be shared across individuals and still move within each at the same rate. The closest work, Meta-AlignNN [10], proposes a method for cross-subject stability; we measure whether a difference in aging rate exists at all, with training-set size held fixed. Appendix A gives the full discussion.

## 3 Data and methods

BraiDyn-BC [6] (DANDI:001425) provides hemodynamically corrected dorsal-cortex $\Delta F/F$ parcellated into 44 Allen Common Coordinate Framework (CCF) regions at $\sim$30 Hz for 25 mice, each completing $\sim$15 daily operant sessions over $\sim$3 weeks. For each onset of four events (lever pull, lick, reward, tone) we build a 44-dimensional evoked feature: mean $\Delta F/F$ over 0–1 s minus the $-0.5$–0 s baseline, per parcel. Each event is decoded against matched negatives by $\ell_2$-regularized logistic regression, scored by area under the receiver-operating characteristic curve (AUC).

Sessions are grouped into an early block (days 1–5) and a late block (days 11–15). The personal decoder leaves out one whole early *session* at a time, training on the mouse's remaining early sessions and scoring on the held-out session (WS) and on the late block (WX). The pooled decoder trains once on the other mice's early blocks and is scored on the same held-out sessions (LS) and late block (LX). Aging is the accuracy lost as testing moves from early to late with training fixed; the per-mouse asymmetry is $(\mathrm{WS} - \mathrm{WX}) - (\mathrm{LS} - \mathrm{LX})$, positive when the personal decoder ages more. Significance is a paired sign-flip permutation test across mice (20,000 permutations), with cluster-bootstrap CIs.

The external validation cohort is Allen Visual Behavior two-photon [1]: containers spanning $\geq$14 days, all experiments in primary visual cortex (VISp) within one 50 µm depth bin (150–199 µm; 25 mice, 184 experiments), leaving 23 mice after per-experiment QC. As different animals have no neuron correspondence and the subset is single-plane, the comparable representation is a permutation-invariant summary of the population response: per-neuron evoked responses described by 25 quantiles plus mean and sd, a fixed 27-dimensional feature. The decoded contrast is `is_change`. Appendix B gives full detail.

Table 1: Personal vs. pooled aging across days (AUC lost from early to late test, holding training fixed), per event, $n = 25$ mice, with whole early sessions held out. Asymmetry = personal − pooled aging; a positive value means the mouse's own decoder ages more. $p$: paired sign-flip permutation. The rightmost column gives the same asymmetry under a trial-level split, the protocol this analysis replaces (Section 4.5). The asymmetry column is the quantity of interest and is positive for every event; the bolded pooled row over all 100 mouse–event pairs is the estimate we defend, while the per-event rows are reported as data rather than as four independent findings. Comparing the last two columns shows how much of each per-event estimate the trial-level split would have added.

| Event | Personal | Pooled | Asymmetry (95% CI) | $p$ | own ages more | trial-split |
|---|---|---|---|---|---|---|
| Lever-pull | $-0.022$ | $-0.025$ | $+0.003$ $[-0.004, 0.011]$ | 0.37 | 64% | $+0.014$ |
| Lick | $+0.012$ | $-0.009$ | $+0.020$ $[0.008, 0.032]$ | 0.0023 | 80% | $+0.028$ |
| Reward | $+0.017$ | $-0.012$ | $+0.029$ $[0.014, 0.046]$ | 0.0005 | 80% | $+0.032$ |
| Tone | $+0.016$ | $+0.001$ | $+0.015$ $[0.001, 0.032]$ | 0.077 | 68% | $+0.022$ |
| **Pooled** | | | $\mathbf{+0.017}$ $[0.010, 0.024]$ | $\mathbf{5 \times 10^{-5}}$ | **73%** | $+0.024$ |

# 4 Results

## 4.1 A population-pooled decoder ages more slowly than a personal one

For each mouse we keep the test data fixed (first the mouse's held-out early sessions, then its late block) and change only the source of the training data: either the mouse itself or the other 24 mice. Across all 100 mouse–event pairs the asymmetry is $+0.017$ AUC ($p = 5 \times 10^{-5}$), with the personal decoder aging more in 73% of pairs (Table 1).

The effect differs among the four events: it is largest for reward ($+0.029$, $p = 0.0005$) and lick ($+0.020$, $p = 0.002$), marginal for tone ($+0.015$, $p = 0.08$), and absent for lever pull ($+0.003$, $p = 0.37$). While we report this variation, we do not interpret it, as the four events all fall within roughly the same two seconds of a trial (Section 4.2). The pooled estimate is the claim we defend; the per-event column is data.

## 4.2 What the four events are

The four decoded events come from a single cued lever-pull trial structure, so they are chained in time rather than independent. Every reward onset has a lever-pull onset a median of $0.083$ s away and a lick onset $0.117$ s away, and $100\%$ and $98\%$ respectively fall inside the window from which a feature vector is built; across all four events the fraction of positive windows containing at least one other event is $100\%$ (reward), $84\%$ (lick), $84\%$ (tone) and $68\%$ (lever pull). The reward decoder is therefore never reading reward in isolation. The negatives are not event-free either: screened only against the event's own onsets, they contain another event 33–$57\%$ of the time. The events also differ tenfold in count, and the per-event asymmetry runs roughly opposite to that count. We cannot separate that explanation from a behavioral one with four events, and do not try; this is why the pooled estimate over all 100 pairs is what we defend (Appendix D).

## 4.3 The personal decoder's advantage erodes over the protocol

The effect also appears as a continuous trend across all fifteen days. The personal decoder starts ahead by about $0.035$ AUC on day 1, and the gap narrows by $0.0014$ AUC per day ($p = 5 \times 10^{-5}$; $65\%$ of pairs), so by day 15 roughly $60\%$ of that advantage is gone. Averaged over events the pooled decoder *gains* over days ($+0.0015$ AUC/day) as the mouse becomes more practiced while the personal decoder stays flat ($+0.0000$ AUC/day): fit to idiosyncrasies, it is unable to ride that wave, and the lift it fails to take is what opens the gap (Appendix C).

## 4.4 The robustness is not a data-volume artifact

A pooled decoder contains $\sim 24\times$ more training events than a personal decoder, so its stability could reflect data volume rather than pooling. We test this with a count-matched control (Fig. 1, left). For each held-out early session $s$ of each mouse we set the training-set size to $n$, the number of events in

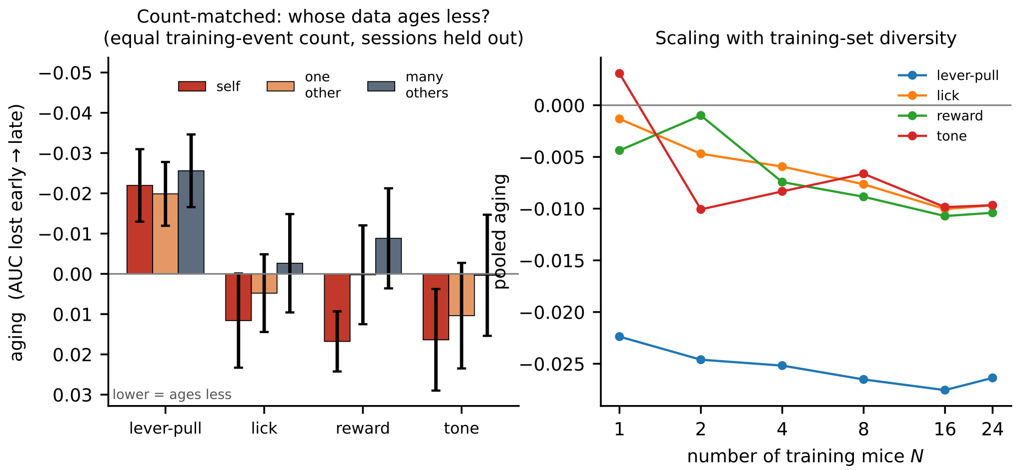


Figure 1: **Left:** count-matched control. With equal numbers of training events, decoder aging (AUC lost from early to late; lower values indicate greater robustness) is compared for the subject's own data (self), one other mouse, and many other mice. **Right:** pooled aging as a function of the number of training mice $N$. Greater diversity provides a small additional gain, with immediate saturation for lever pull. The reader should take two things from this panel pair: at equal training-event count the *self* bar ages most for three of the four events, so the effect is not explained by data volume; and *many* ages least for all four, so diversity of training individuals adds a further, smaller benefit.

that mouse's remaining early sessions, and compare three decoders: *self*, trained on those $n$ events; *one*, trained on $n$ events from a single other mouse; and *many*, trained on $n$ events drawn across the other 24 mice. All three are scored on the same held-out session $s$ and the same late block, so they differ only in whose data they were fit on.

For the events where there was an asymmetry, the count-matched ordering reproduces it. The self decoder ages more than a decoder trained on one other mouse by $+0.010$ AUC on average, and by $+0.007$ for lick, $+0.017$ for reward and $+0.006$ for tone individually. As both decoders see the same number of training events, this is not a data-volume artifact.

The diversity ordering is the most robust result in this analysis: *many* ages less than *one* for all four events, including lever pull ($-0.026$ vs. $-0.020$). Varying the number of training mice $N \in \{1, \ldots, 24\}$ (Fig. 1, right) confirms this is a secondary effect of around $0.01$ AUC, comparable to the self-versus-one-other gap: most of the resistance to drift is already present when training on a single other individual, and additional individuals refine it further.

### 4.5 How much the validation protocol matters

Every number above holds out whole sessions. The identical analysis on identical features with a trial-level split inflates the personal decoder's within-block accuracy by $+0.013$ (lever pull), $+0.010$ (lick), $+0.009$ (reward) and $+0.000$ AUC (tone), while leaving the pooled decoder unaffected – it is trained on other animals and shares no session with the test data. The inflation therefore lands on one side of the difference (rightmost column of Table 1), and for lever pull it is the entire effect: a significant-looking $+0.014$ ($p = 0.001$) becomes $+0.003$ ($p = 0.37$). Any comparison of a within-subject decoder against a cross-subject one is exposed to this, since only the within-subject arm can share a session between train and test (Appendix E).

### 4.6 The effect replicates on an independent dataset

To further test whether the asymmetry is a property of that one cohort, modality and task, we repeat the analysis on the Allen Visual Behavior two-photon dataset [1]: a different laboratory, cellular two-photon rather than mesoscale widefield imaging, a visual change-detection task rather than a cued lever pull, and a 27-dimensional permutation-invariant population summary rather than a 44-parcel

Table 2: The aging asymmetry on the original cohort and on the external validation dataset. Aging is $\mathrm{AUC}$ lost from early to late test with training held fixed; asymmetry is personal $-$ pooled aging, positive when the subject's own decoder ages more. Both rows use the same session-held-out protocol and the same count-matched control. $p$: paired sign-flip permutation. The two rows are independent cohorts differing in laboratory, recording modality, task and feature space; both asymmetries are positive and of comparable magnitude, which is the replication claim. The note below shows the count-matched control reproducing the same ordering in both.

| | Personal | Pooled | Asymmetry (95% CI) | $p$ | own ages more |
|---|---|---|---|---|---|
| BraiDyn-BC, widefield, 25 mice | | | $+0.017$ $[0.010, 0.024]$ | $5 \times 10^{-5}$ | 73% |
| Allen VB, two-photon, 23 mice | $+0.024$ | $+0.001$ | $+0.023$ $[0.010, 0.037]$ | 0.003 | 74% |

*Count-matched control, aging at equal training-event count.* BraiDyn-BC: self $-$ one other mouse $= +0.010$; many others age less than one on all four events. Allen VB: self $+0.024$, one $+0.013$, many $+0.003$; self $-$ one $= +0.011$, many $-$ one $= -0.010$.

atlas feature. The decoded contrast is a stimulus change, so it is much less coupled to reward and licking than anything decoded on BraiDyn-BC.

The asymmetry reproduces at a comparable magnitude (Table 2). Across 23 mice the personal decoder ages by $+0.024$ $\mathrm{AUC}$ and the pooled decoder by $+0.001$, an asymmetry of $+0.023$ $[0.010, 0.037]$, $p = 0.003$, with the mouse's own decoder ageing more in 74% of animals. Early-block accuracy is $0.672$ for the personal decoder and $0.638$ for the pooled one, so the familiar ordering holds at the start; by the late block the personal decoder leads by only $0.012$, about a third of its initial advantage.

The mechanism identified is not identical in both datasets; on BraiDyn-BC the gap narrows mostly because the pooled decoder *gains* over days as the animal becomes more practised (Section 4.3), while the personal decoder stays flat. Here the pooled decoder is flat ($+0.001$) and the entire asymmetry comes from the personal decoder losing accuracy. The shared finding within both, hence, is that the personal decoder ages faster.

The count-matched control reproduces as well. At equal training-event count the self decoder ages by $+0.024$, a decoder from one other mouse by $+0.013$ and one drawn across the other mice by $+0.003$. The self-minus-one gap is $+0.011$ $[0.001, 0.022]$, close to the $+0.010$ measured on BraiDyn-BC, though on 23 animals it is marginal by the sign-flip test ($p = 0.054$); self-minus-many is $+0.021$ $[0.010, 0.033]$, $p = 0.002$. The diversity ordering again holds: many other mice age less than one by $-0.010$ $[-0.016, -0.004]$, $p = 0.003$. The result also does not depend on where the inclusion threshold is drawn, giving $+0.022$ at three or more sessions per mouse (24 mice), $+0.023$ at four (23 mice), $+0.023$ at five (22 mice) and $+0.024$ at six (18 mice).

One thing does not transfer. Repeating the trial-level split here *lowers* the asymmetry to $+0.014$ ($p = 0.07$) rather than raising it. Separating the two differences on the same trials: trial shuffling still inflates the personal decoder by $+0.015$, as on BraiDyn-BC, but scoring one pooled $\mathrm{AUC}$ across sessions instead of averaging per-session values costs $-0.030$ ($p < 10^{-4}$) and dominates. The corrected estimate is here the *larger* one, so the replication is not an artifact of the split.

## 5 Discussion

A subject's own decoder overfits idiosyncratic features of its early sessions, some of which later drift. A decoder that has never seen the subject must instead rely on components of the code that are shared across individuals and more stable over time. The count-matched control supports this reading directly: at equal data, other-individual data ages less than the subject's own. Unlike within-subject recalibration, this requires no newly labelled data from the individual at test time – which is the practical point, since obtaining that data is the expensive part of deploying a decoder over months.

The effect is small in absolute terms, and we think it is worth being explicit about why we believe it anyway. It rests on 25 animals and 357 recording sessions, and it is supported by five checks that are not variations on one analysis. A two-block contrast and a day-by-day regression are different estimators and select the same events. The count-matched control fixes the training-set size, so the result cannot be a data-volume artifact. Replacing the linear readout with a one-dimensional convolutional neural network (CNN) or a gated recurrent unit (GRU) leaves it in place. Holding out whole sessions rather than trials removes an inflation large enough to eliminate one of the four events entirely – the effect reported here is what remains after that correction, not before it. And it reproduced in a second cohort recorded in another laboratory with a different instrument and a different task. Each of these could have removed the result and none did.

**Limitations.** The clearest limitation is what the four events are. They are not four independent behaviors but four thresholded channels of one cued lever-pull trial, overlapping to the point where $100\%$ of reward windows also contain licking, and differing tenfold in event count. We therefore cannot always say which behavior the effect belongs to. We also measure at the mesoscale, using parcel-averaged activity, and the analysis covers a two-week interval in one species. The two-photon replication reaches single neurons only at one depth and one visual area. Finally, the trial-level inflation of Section 4.5 is a caution as much as a result: it is large enough to manufacture an effect that is not there. Both datasets are public (DANDI:001425; Allen Brain Observatory) and we release the analysis code and streaming feature pipeline, specifying all splits, seeds, and hyperparameters.

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# A Extended related work

**Representational drift and its mitigation.** Drift is characterized within individual subjects [18, 3, 20, 16], and most approaches to mitigating it also operate within the subject. These include manifold stabilization [2], supervised and self-calibrating recalibration [5, 15], and adversarial or nonlinear alignment to the subject's own reference day [4, 14]. Each method uses later data from the same subject to stabilize the decoder. We instead ask whether data from other subjects can provide that stability.

**Cross-subject decoding.** Shared structure across individuals already supports cross-subject readout through manifold alignment [9], contrastive embeddings [19], and, on this cohort, atlas-aligned foundation models [21]. Multi-participant pooling can also reduce overfitting [13, 12]. Our question differs in two ways. Primarily, prior work measures accuracy or transfer at a fixed time, while we look at how that accuracy changes over days, to see whether pooled decoders age more slowly. Further, we keep the training-set size fixed with the count-matched control, hence being able to delegate the relevant overfitting not to limited data but to idiosyncratic features that cause drift.

The closest work to ours combines both axes: Meta-AlignNN [10] uses the consistency of neural population dynamics across individuals to build a decoder that remains stable across subjects, time and tasks. Meta-AlignNN proposes a method for achieving stability; we measure whether a difference in aging rate exists at all between a decoder trained on the subject and one trained on others, with the training-set size held fixed. A measured effect of this kind is what would justify the design choice such methods already make.

**Conservation of cortex-wide dynamics.** Cortex-wide activity contains shared, low-dimensional, movement-related components [11, 7, 8], and latent dynamics are preserved across animals performing similar behaviour to the point that a decoder fit to one animal transfers to another [17]. That conservation is the mechanism a pooled decoder would exploit, and it is why pooling is possible at all here. It does not, however, imply that pooling will *resist drift*: a code can be shared across individuals and still move within each of them at the same rate.

# B Full data and methods

**Dataset and features.** BraiDyn-BC [6] (DANDI:001425, CC-BY) provides, for each mouse and session, hemodynamically corrected dorsal-cortex $\Delta F/F$ traces parcellated into 44 Allen Common Coordinate Framework regions at $\sim$30 Hz, together with time-aligned behavioral channels: 25 mice, each completing $\sim$15 daily operant sessions over $\sim$3 weeks. We detect the onset of four events (lever pull, lick, reward, and tone) as rising edges, with a 1 s refractory period. For each onset we construct a 44-dimensional evoked feature by subtracting the mean pre-onset baseline ($-0.5$–$0$ s) from the mean post-onset $\Delta F/F$ response (0–1 s) in each parcel. Negative samples are drawn from matched-count windows more than 2 s from any onset *of the same event*; they are not screened against the other three events, and Section 4.2 reports how often they contain one. Every classification is a binary decode of one event against these matched negatives, scored by area under the ROC curve ($\mathrm{AUC}$). The four decoders are therefore four independent binary detectors, not a four-way classifier.

**Decoders.** The primary decoder is a standardized $\ell_2$-regularized logistic regression on the 44-dimensional evoked feature, the natural comparison to prior cross-subject decoders, which are overwhelmingly linear [9, 13]. Every condition below uses this same readout, so the only variable is which data the decoder was trained on.[1]

**Temporal blocks and aging.** We group each mouse's sessions into an early block (days 1–5) and a late block (days 11–15). The personal decoder is estimated by leaving out one whole early *session* at a time: for each early session $s$ we train on the mouse's remaining early sessions and score the resulting decoder both on $s$ (within-mouse same-block, WS) and on the late block (within-mouse across-block, WX), then average over $s$. The pooled decoder is trained once on the other mice's early blocks and scored on the same held-out sessions $s$ (LS) and on the late block (LX). Aging is the accuracy lost when testing moves from the held-out mouse's early data to its late block while training stays fixed: personal aging is $\mathrm{WS} - \mathrm{WX}$ and pooled aging is $\mathrm{LS} - \mathrm{LX}$. The per-mouse aging asymmetry is $(\mathrm{WS} - \mathrm{WX}) - (\mathrm{LS} - \mathrm{LX})$; a positive value means the personal decoder ages more. We assess significance with a paired sign-flip permutation test across mice (20,000 permutations) and compute 95% confidence intervals with a cluster bootstrap over mice.

**External validation dataset.** Section 4.6 repeats the analysis on the Allen Visual Behavior two-photon dataset [1]. We select containers spanning at least 14 days whose experiments are all in VISp within a single 50 µm depth bin, which is the cross-subject correspondence available here; the largest such band, 150–199 µm, holds 25 mice and 184 experiments. We require at least 15 segmented cells and 40 usable trials per experiment, leaving 166 experiments and 23 mice with at least four sessions. As different animals have no neuron correspondence and the longitudinal subset is single-plane, averaging over the field of view would give a one-dimensional feature. The representation that is comparable across animals is therefore a permutation-invariant summary of the population response: per neuron we take the mean $\Delta F/F$ over 0–1 s minus the mean over $-0.5$–0 s, then describe the distribution of those evoked responses by 25 quantiles plus its mean and standard deviation, giving a fixed 27-dimensional feature regardless of how many cells were segmented. The decoded contrast is `is_change`, a real image change against a sham catch trial, which is a stimulus contrast rather than a reward- or lick-coupled one.

# C Day-by-day decay regression

The effect also appears as a continuous trend across all fifteen days. For each session day we compute the accuracy of the personal decoder and the pooled decoder and regress each on day; on early days the personal decoder is trained by leave-one-session-out, so it is never tested on a session it trained on. The personal decoder starts ahead, leading by about $0.035$ AUC on day 1. That lead erodes: the gap narrows by $0.0014$ AUC per day ($p = 5 \times 10^{-5}$; narrowing in 65% of pairs), so by day 15 roughly 60% of the early-week advantage is gone. Averaged over the four events, the pooled decoder gains accuracy over days ($+0.0015$ AUC/day) as the sessions become easier – the mouse is more practiced – while the personal decoder stays flat ($+0.0000$ AUC/day). The personal decoder, fit to idiosyncrasies, is unable to ride that wave, and the lift it fails to take is what opens the gap.

# D Event timing and overlap

The four decoded events come from a single cued lever-pull trial structure, so they are chained in time rather than independent. Every reward onset has a lever-pull onset a median of $0.083$ s away and a lick onset $0.117$ s away, and 100% and 98% respectively fall inside the window from which a feature vector is built. Across all four events, the fraction of positive windows containing at least one other event is 100% (reward), 84% (lick), 84% (tone) and 68% (lever pull). The reward decoder is therefore never reading reward in isolation; its input window always contains licking and lever movement as well. The negatives are not event-free either: screened only against the event's own onsets, they contain another event 33–57% of the time.

The events also differ heavily in sample size, from a median of 56 reward onsets per session against 107 (tone), 530 (lick) and 652 (lever pull). The per-event asymmetry runs roughly opposite to that count; a decoder fit to fewer events has more idiosyncrasy available to overfit, and hence more for

[1]The result is not a property of the linear readout. Repeating the analysis with a 1-D convolutional network over time and a GRU over the frame sequence, under the same session-held-out protocol, the asymmetry survives for both events that carry it: lick gives $+0.020$, $+0.012$ and $+0.010$ and reward $+0.029$, $+0.015$ and $+0.012$ under the linear, CNN and GRU readouts, all significant.

pooling to avoid. We cannot separate that explanation from a behavioral one with four events, and we do not try. This is why the pooled estimate over all 100 pairs is what we defend.

## E Validation protocol in full

Every number above holds out whole sessions. Repeating the identical analysis on the identical features with a trial-level split – shuffling trials pooled across the early sessions, so that a decoder may train on one part of a session and be tested on another – inflates the personal decoder's within-block accuracy by $+0.013$ (lever pull), $+0.010$ (lick), $+0.009$ (reward) and $+0.000$ AUC (tone). The pooled decoder is unaffected, because it is trained on other animals and has no session in common with the test data.

The inflation therefore lands on one side of a difference between two arms (rightmost column of Table 1). For lever pull it is the entire effect: a significant-looking $+0.014$ ($p = 0.001$) becomes $+0.003$ ($p = 0.37$). We report this because the pattern generalizes beyond our dataset. Any comparison of a within-subject decoder against a cross-subject one is exposed to it, since only the within-subject arm can share a session between train and test.